Regular expressions

Patterns for searching, validating and replacing text

Yarrowpine Developer Cheatsheets

CHARACTERS

`.`	any character except a line break
`\d \w \s`	digit, word character, whitespace
`\D \W \S`	the opposite of each of the above
`[abc] [a-z] [^0-9]`	one of a set, a range, anything but
`\. \* \?`	escape a special character to match it

ANCHORS

`^ $`	start and end of the line or string
`\b`	word boundary: \bcat\b skips "concat"
`\A \z`	absolute start and end (most engines)

QUANTIFIERS

`a* a+ a?`	zero or more, one or more, optional
`a{3} a{2,} a{2,4}`	exactly 3, at least 2, between 2 and 4
`a+? a*?`	lazy: match as little as possible

GROUPS

`(abc)`	capture group, reused as \1 or $1
`(?:abc)`	group without capturing
`(?<year>\d{4})`	named group
`cat|dog`	either alternative

LOOKAROUND

`x(?=y)`	x only if followed by y
`x(?!y)`	x only if not followed by y
`(?<=y)x`	x only if preceded by y
`(?<!y)x`	x only if not preceded by y

FLAGS

`i`	ignore case
`m`	^ and $ match at every line
`s`	. also matches line breaks
`g`	find all matches, not just the first
`x`	allow spaces and comments in the pattern

RECIPES

`^\d{4}-\d{2}-\d{2}$`	date shaped like 2026-11-04
`^\S+@\S+\.\S+$`	rough email shape check
`^#?[0-9a-fA-F]{6}$`	six-digit hex colour
`\b(\w+)\s+\1\b`	a word typed twice in a row
`^\s+|\s+$`	leading or trailing spaces (to trim)
`(?<=order-)\d+`	the number in "order-1042"

GOOD HABITS

- Engines differ: test patterns in the language you will use.
- Shape checks are not full validation; confirm emails by sending a link.
- Avoid nested quantifiers like (a+)+ on user input; they can be very slow.

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